

Repairing the Round-8 Completeness Construction for $\mathcal{ALCI}_{\text{RCC5}}$: Composition-Forced Verticalization

Independent repair note (Claude, Anthropic), prompted by Michael Wessel

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Abstract

The companion review ([review3/review_round8_split_forest.pdf](#)) identified a gap (D-1) in the completeness direction of GPT-5.5’s round-8 saturated-side-context proof: a DR/PO side witness can be RCC5-composition forced to be a proper part of an *infinite* pumped containment tower, and the bounded saturated side context cannot record the forced PP/PPI labels on the tower’s tail, so the certificate produced by the stated construction is invalid. This note supplies a repair. The repair rests on one principle — the vertical backbone must present the *full* generated containment order, not merely the selected-witness edges — realised by a single new operation (splicing a composition-forced witness into the cover relation $E_{\uparrow}$ at a bounded height). We prove the three facts the repair needs: (A) the forced label up an ascending chain is monotone and eventually constant (*threshold* lemma, machine-checked); (B) the threshold is bounded by the number of boundary descriptors, so finiteness of the certificate universe is preserved; and (C/D) the repaired presentation embeds label-preservingly into the fixed satisfying model, so RCC5 composition and universal safety are inherited rather than re-derived, and the descriptor space stays finite. We restate the patched split-cover lemma and the restored completeness theorem, re-derive the stress concept C'_0 (now certifiable) and its UNSAT sibling (still correctly rejected), and close with an honest list of the obligations a referee should still check. All composition arithmetic and the two worked certificates are machine-checked by the accompanying scripts.

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1 Purpose and summary

We work entirely inside the framework of the round-8 expanded companion `expanded_split_forest_full_details`. definitions and validity clauses are cited by their names there (“thinning”, “split-cover”, “eq-aware-vertical”, “incidence-tags”, “pair-shape”, “side-width”, “replacement”, “blocking”, and the validity conditions (V1)–(V15)). Nothing in `alcirc5/` is modified; this is a standalone replacement for the witness-placement step of the completeness proof.

The gap, in one sentence: the round-8 construction places every DR/PO witness w of a source u inside a *bounded* saturated side context $S_k(u)$, but RCC5 composition can force w to be PP/PPI-comparable with *unboundedly many* vertical ancestors of u , and a bounded side context cannot carry those forced labels (companion review, Proposition “fail”).

The repair, in one sentence: do not file such a w under u ’s side context; instead detect the composition-forced comparability and *splice* w into the vertical cover relation $E_{\uparrow}$ at the (provably bounded) height where it first becomes a proper part, so that the equality-aware reachability Reach_{PP} — which already scales to infinite towers — carries the correct PP/PPI label to the entire tail.

The conceptual reason the repair is clean: the completeness construction is *presenting a fixed, already-consistent model* $\mathcal{J}$, not synthesising a new one. The bug was presentational (the forest had no slot for a composition-forced vertical pair), not a failure of consistency. A presentation that routes the pair correctly inherits RCC5 composition and universal safety *from* $\mathcal{J}$ for free.

2 The generated order and the presentation principle

Fix a witness-generated model $\mathcal{J} = (W, \rho)$ obtained from the thinning lemma, with W countable and every closure existential witnessed inside W . Write $x < y$ for $\rho(x, y) = \text{PP}$ (“ x is a proper part of y ”).

Proposition 2.1 (The generated containment order). *($W, <$) is a strict partial order: irreflexive, transitive, and antisymmetric.*

Proof. Irreflexivity: $\rho(x, x) = \text{EQ} \neq \text{PP}$ by strong equality. Transitivity: if $x < y < z$ then $\rho(x, z) \in \text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$, so $x < z$. Antisymmetry: if $x < y$ and $y < x$ then $\rho(x, y) = \text{PP}$ and $\rho(y, x) = \text{PP}$; but $\rho(y, x) = \text{inv}(\rho(x, y)) = \text{PPI} \neq \text{PP}$, a contradiction. (All three use only the frame axioms and $\text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$.) $\square$

Principle 2.2 (Faithful backbone). *The vertical backbone of the split presentation must generate, under equality-aware transitive closure, the entire restriction of $<$ to the generated occurrences, not merely the sub-relation given by selected PP/PPI witness edges.*

The round-8 construction violates Principle 2.2: it builds $E_{\uparrow}$ only from selected PP/PPI witnesses, so a generated element w that happens (by composition) to satisfy $w < a$ for some ancestor a —

without a having been *selected* as a PP-witness of w — has no $E_\uparrow$ -path to a , and the pair (a, w) is left to the residual frontier, where the label PP/PPI is illegal. The repair restores Principle 2.2 by adding exactly the missing cover edges, at bounded height.

3 Corrected witness placement

3.1 Where composition forces verticality

Definition 3.1 (Ancestor profile of a side pair). *Let $u \in W$ have a selected DR/PO witness w (so $\rho(u, w) \in \{\text{DR}, \text{PO}\}$), and let a be a $<$ -ancestor of u (so $u < a$, i.e. $\rho(a, u) = \text{PPI}$). The ancestor relation of the pair at a is $r_a := \rho(a, w)$.*

Lemma 3.2 (Forced range). *In the situation of Definition 3.1, $r_a \in \{\text{DR}, \text{PO}, \text{PPI}\}$; in particular $r_a \notin \{\text{EQ}, \text{PP}\}$.*

Proof. $r_a \neq \text{EQ}$: else $a = w$, but $\rho(a, u) = \text{PPI}$ while $\rho(w, u) = \text{inv}(\rho(u, w)) \in \{\text{DR}, \text{PO}\}$. $r_a \neq \text{PP}$: $\rho(a, w) = \text{PP}$ means $a < w$; with $u < a$ this gives $u < w$ by Proposition 2.1, so $\rho(u, w) = \text{PP}$, contradicting $\rho(u, w) \in \{\text{DR}, \text{PO}\}$. The remaining base labels are DR, PO, PPI. $\square$

Thus a side pair is comparable with an ancestor exactly when $r_a = \text{PPI}$ (then $w < a$, w a proper part of a). We must route w into the backbone precisely on the set $\{a : r_a = \text{PPI}\}$.

3.2 The splice

Definition 3.3 (Corrected split-cover with splicing). *Construct the weak split presentation as in companion Lemma “split-cover”, with one change. When a selected DR/PO witness w of u is generated, also inspect the ancestor relations r_a for the $<$ -ancestors a of u . Let*

$$A_{\text{PPI}}(w) = \{a : u < a \text{ and } r_a = \text{PPI}\} = \{a : w < a\}.$$

If $A_{\text{PPI}}(w) \neq \emptyset$, let a^ be a $<$ -minimal element of $A_{\text{PPI}}(w)$ (it exists and is unique on the ancestor chain of u , by Lemma 4.1 below). Add the cover edge $w E_\uparrow a^*$ (the splice), so that w becomes a vertical proper part of a^* and, by transitivity of $E_\uparrow$, of every $a \in A_{\text{PPI}}(w)$. Demote the pair (u, w) and every pair (a, w) with $a \notin A_{\text{PPI}}(w)$ to its inherited $\{\text{DR}, \text{PO}\}$ residual/side label. If $A_{\text{PPI}}(w) = \emptyset$, leave w as a pure side witness exactly as in round-8.*

The splice edge $w E_\uparrow a^*$ asserts the genuine fact $w < a^*$ of $\mathcal{J}$, so it is sound by construction; the next sections show it is well-defined, bounded, and globally consistent.

4 The threshold lemmas

4.1 Monotonicity

Lemma 4.1 (Monotone threshold). *Let w be a DR/PO witness of u , and let $a_1 < a_2 < a_3 < \dots$ be the $<$ -ancestor chain of u (so $u < a_1$ and $a_m < a_{m+1}$). Put $r_m = \rho(a_m, w)$. Then, with respect to the linear order $\text{DR} \succ \text{PO} \succ \text{PPI}$, the sequence (r_m) is non-increasing:*

$$r_{m+1} \in \text{Comp}(\text{PPI}, r_m) \quad \text{and} \quad r_{m+1} \preceq r_m.$$

Consequently (r_m) is eventually constant; there are thresholds $1 \leq k_1 \leq k_2 \leq \infty$ with $r_m = \text{DR}$ for $m < k_1$, $r_m = \text{PO}$ for $k_1 \leq m < k_2$, and $r_m = \text{PPI}$ for $m \geq k_2$. In particular $\{a : w < a\}$ is an upward-closed final segment of the chain with $<$ -minimum a_{k_2} (when $k_2 < \infty$).

Proof. $\rho(a_{m+1}, a_m) = \text{PPI}$ since $a_m < a_{m+1}$, so $r_{m+1} \in \text{Comp}(\rho(a_{m+1}, a_m), \rho(a_m, w)) = \text{Comp}(\text{PPI}, r_m)$. By Lemma 3.2 each $r_m \in \{\text{DR}, \text{PO}, \text{PPI}\}$, and the relevant transitions are

$$\text{Comp}(\text{PPI}, \text{DR}) = \{\text{DR}, \text{PO}, \text{PPI}\}, \quad \text{Comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}, \quad \text{Comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}.$$

Inspection shows no transition increases the rank in $\text{DR} \succ \text{PO} \succ \text{PPI}$: from DR one stays $\leq \text{DR}$; from PO the options $\{\text{PO}, \text{PPI}\}$ are $\leq \text{PO}$ (note $\text{DR} \notin \text{Comp}(\text{PPI}, \text{PO})$, so no return to DR); from PPI the only option is PPI. A non-increasing sequence in a three-element order is eventually constant, giving the thresholds. Upward-closure of $\{a : w < a\}$ is PPI-absorption ($\text{Comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$). $\square$

All three transition entries, the non-increase, and the PPI-absorption are machine-checked in `fix_feasibility.py`.

Remark 4.2 (The benign regimes). *If $k_2 = \infty$ the witness is never a proper part of an ancestor; then every (a_m, w) carries an inherited DR or PO label, which is a legal residual/side shape, and (V12) holds because the labels are inherited from the model $\mathcal{J}$. There is no gap in these regimes; the round-8 construction already handles them. The repair is needed only when $k_2 < \infty$.*

4.2 Boundedness

For finiteness we must bound k_2 . Recall the boundary descriptor of companion Definition “boundary-descriptor” and the replacement/blocking lemmas (“replacement”, “blocking”); a checkpoint of the chain is a generated component, and there are at most N distinct boundary descriptors for fixed C_0 , where $N \leq 2^{\text{poly}(n)}$ is the round-8 descriptor bound.

Lemma 4.3 (Bounded threshold). *The chain may be re-presented, preserving all closure formulas and finite-prefix RCC5 composition, so that the splice height satisfies $k_2 \leq 3N$.*

Proof. By Lemma 4.1 the chain splits into three consecutive constant- r stretches: a DR-stretch $[1, k_1)$, a PO-stretch $[k_1, k_2)$, and the PPI-tail $[k_2, \infty)$. Consider any one stretch. Along it the relation r_m to the fixed occurrence w is constant, so the value of r_m is not part of what distinguishes checkpoints within the stretch; the only distinguishing data is the boundary descriptor, of which there are at most N . If a stretch has more than N checkpoints, two of them, say at heights $i < j$, have equal boundary descriptors. By the replacement lemma the segment $[i, j)$ may be excised and a_{i-1} glued directly to a_j , preserving all closure formulas and finite-prefix composition. Because r is constant on the stretch, $r_{i-1} = r_j$, so the glue is coherent with the relation to w as well: the inherited label between w and the glued checkpoint is unchanged. Iterating, each stretch is shortened to at most N checkpoints, so $k_1 \leq N$ and $k_2 \leq 2N < 3N$. (The bound $3N$ leaves room for the single glued boundary checkpoints between stretches.) The PPI-tail is then a standard blocking region: $r = \text{PPI}$ is constant, and ordinary blocking applies. $\square$

Remark 4.4 (Periodic witness families). *Lemma 4.3 bounds the splice height for a single fixed witness w . When a whole periodic family of witnesses occurs — e.g. every tower node carries its own DR/PO witness that becomes its proper part — the splice is not placed in the prefix but is incorporated into the recurrent cyclic profile itself (“this profile splices a proper-part side child here”), and pumping reproduces one fresh spliced witness per lap. Cross-lap pairs between these fresh witnesses are ordinary occurrence-sensitive pairs. Either way the certificate stays finite.*

5 Consistency and finiteness are inherited from $\mathcal{J}$

Lemma 5.1 (Acyclicity of the spliced backbone). *The equality-aware transitive closure Reach_{PP} of the spliced cover relation $E_{\uparrow}$ is a strict partial order; in particular the splice introduces no containment cycle.*

Proof. Every $E_{\uparrow}$ edge — selected or spliced — asserts a genuine relation $x < y$ of $\mathcal{J}$ (the splice edge $w E_{\uparrow} a^*$ asserts $w < a^*$, which holds because $a^* \in A_{\text{PPI}}(w)$). Equality mates have, by construction, identical $<$ -relations to all occurrences (typed congruence). Hence $\text{Reach}_{\text{PP}} \subseteq <$ on origins, and $<$ is a strict partial order by Proposition 2.1. A sub-relation of a strict partial order is acyclic. $\square$

Lemma 5.2 (Label-preserving embedding). *The quotient of the corrected presentation by explicit equality maps label-preservingly onto the generated submodel of $\mathcal{J}$. Consequently every RCC5-composition obligation and every universal-safety obligation of the certificate holds, because it holds in $\mathcal{J}$.*

Proof. Each occurrence has an origin in $\mathbf{W}$; equality mates share an origin; the splice changes only *which presentation device* carries a pair’s label (a vertical up/down shape instead of an impossible residual shape), never the label itself, which is inherited from ρ . After the splice, every pair (a, w) with $w < a$ has $\text{Reach}_{\text{PP}}(w, a)$ (Definition 3.3), hence the down shape with label $\text{PPI} = \rho(a, w)$; every pair with $r_a \in \{\text{DR}, \text{PO}\}$ keeps its inherited residual/side label. Thus the map origin: occurrence $\mapsto \mathbf{W}$ is a label isomorphism onto the generated submodel. Composition (V8) for any triple of occurrences is the composition of their origins in $\mathcal{J}$; universal safety (V10)/(V12) for any pair is the universal it satisfies in $\mathcal{J}$. Both transfer. $\square$

Lemma 5.3 (Descriptor space stays finite). *The repair does not enlarge the descriptor universe beyond a finite bound.*

Proof. A boundary descriptor records reachable *types* (subsets of $\text{Cl}(C_0)$, $\leq 2^n$ of them) and side-mosaic *isomorphism types* (finitely many by the side-width bound), not individual occurrences. Splicing adds, to a checkpoint’s data, at most the finite list of incident splice-cover edges; each such edge is described by the (bounded) profile/port of its endpoints and one of five labels. In the recurrent tail the relation to spliced witnesses is the constant PPI and contributes only the presence of the witness type in the downward type-set — already a bounded datum. Hence the descriptor space is multiplied by a bounded factor and remains finite; the round-8 finiteness counts (Lemmas “side-width”, “finite-universe”) are preserved up to that factor. $\square$

6 Compatibility with split copies (overlap)

A side witness w may be forced to share a proper part with its source.

Lemma 6.1 (Overlap reduces to split copies). *Let w be a DR/PO witness of u and $d < u$ a generated descendant. Then $\rho(w, d) \in \text{Comp}(\rho(w, u), \rho(u, d)) = \text{Comp}(\{\text{DR}, \text{PO}\}, \text{PPI})$; in the PO case $\rho(w, d) \in \{\text{DR}, \text{PO}, \text{PPI}\}$. If $\rho(w, d) = \text{PPI}$ (so $d < w$ as well as $d < u$), then d is a common proper part of the PO-incomparable elements u and w , and is represented by the existing split-copy discipline: occurrences $d^{(1)} E_{\uparrow}$ -below u and $d^{(2)} E_{\uparrow}$ -below w with $d^{(1)} \equiv_{\text{EQ}} d^{(2)}$.*

Proof. The composition values are read off the round-8 table ($\text{Comp}(\text{PO}, \text{PPI}) = \{\text{DR}, \text{PO}, \text{PPI}\}$, $\text{Comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$; machine-checked). When $\rho(w, d) = \text{PPI}$, $d < u$ and $d < w$ hold simultaneously while $\rho(u, w) \in \{\text{DR}, \text{PO}\}$, which is exactly the configuration the round-8 split-copy discipline

(companion Lemma “multiple-superpart-split”) was designed for: a single origin d placed under two $<$ -incomparable parents via equality-mate occurrences. Typed congruence makes the two copies carry identical labels to all third occurrences, and quotienting recovers the single origin d . The set of such common parts is a subset of the generated descendants of u , finite per generated component, so no new infinity is introduced. $\square$

7 Patched statements

Definition 7.1 (Patched incidence of spliced pairs). *In companion Definition “incidence-tags”, a pair (u, v) that is Reach_{PP} -related through a splice edge receives the vertical tag up (resp. down) and hence the mandatory label PP (resp. PPI), exactly as for selected vertical pairs. The tag side_R is reserved for pairs inside one bounded saturated context; it is not used for a PP/PPI pair that the splice has placed in Reach_{PP} . This removes the round-8 ambiguity (review3 §4) by declaring that verticalizing a forced pair contributes to the global Reach_{PP} , not merely to a local mosaic.*

Lemma 7.2 (Patched split-cover). *Definition 3.3 produces a weak split presentation whose strong quotient is label-isomorphic to the generated submodel of $\mathcal{J}$, in which every pair with origin label PP/PPI is realised by equality-aware vertical reachability (selected or spliced), every EQ pair by explicit equality, and every residual/side pair by an inherited DR/PO label.*

Proof. Round-8 Lemma “split-cover” already realises selected PP/PPI witnesses, EQ mates, and DR/PO side witnesses. The only pairs it failed to realise were (a, w) with $w < a$ for a DR/PO witness w and a non-selected ancestor a (the D-1 pairs). By Definition 3.3 the splice makes $\text{Reach}_{\text{PP}}(w, a)$ hold for every such a (Lemma 4.1: $\{a : w < a\}$ is the final segment above a^* , and $a^* E_{\uparrow}$ -reaches all of it), so these pairs now carry the down shape with the correct label PPI. Acyclicity is Lemma 5.1; label-preservation is Lemma 5.2; overlap is Lemma 6.1. Hence every origin PP/PPI pair is vertical, as claimed. $\square$

Remark 7.3 (Validity clauses). *No validity clause is weakened. The only structural change visible to the checker is that some pairs previously destined for the residual frontier are now up/down pairs; (V6) pair-shape determinacy, (V8) composition, and (V10) vertical universal safety apply to them unchanged. Lemma 4.3 keeps every bound in (V15) finite. Optionally one adds a single bookkeeping clause:*

(V16) Forced verticalization. *For every DR/PO witness w of a source u and every ancestor a of u with $\rho(a, w) = \text{PPI}$ in the recorded labels, $w \text{ Reach}_{\text{PP}} a$ holds.*

This clause is decidable (a finite reachability test) and is exactly what the splice guarantees. It turns the D-1 obstruction into a checkable condition.

8 Completeness restored

Theorem 8.1 (Completeness, repaired). *If C_0 is satisfiable, then C_0 has a valid finite generated split-forest certificate (in the sense of the round-8 companion, augmented by Definition 7.1 and optionally (V16)).*

Proof. Let $\mathcal{J}, a_0 \models C_0$ be witness-generated (thinning lemma). Build the corrected split presentation (Definition 3.3); it is a faithful label-preserving presentation of the generated submodel (Lemma 7.2). Extract a finite certificate by the round-8 extraction procedure (companion Lemma “extraction”), with two changes: (i) splice edges are recorded as cover edges of $E_{\uparrow}$, contributing to the up/down

pair shapes; (ii) per-witness splice heights are normalised to $\leq 3N$ by Lemma 4.3, or absorbed into recurrent profiles by Remark 4.4. All validity clauses hold: type legality and witness support from satisfaction in $\mathcal{J}$; equality congruence from equality of origins; RCC5 composition and universal safety by Lemma 5.2; request-closure and universal safety on cycles from the blocking lemma applied to the PPI-tail and the bounded prefix; finiteness from Lemmas 4.3 and 5.3. The extracted object is therefore a valid certificate. $\square$

Together with round-8 soundness (untouched by this note) and the finite, effective enumeration (preserved by Lemma 5.3), this restores the decision procedure and hence Theorem “main” of the round-8 companion.

9 Worked cases

9.1 C'_0 : a valid certificate now exists

Recall $G = (\forall \text{PO}.X) \sqcap (\exists \text{PP}.G)$ and $C'_0 = (\exists \text{PP}.G) \sqcap (\exists \text{PO}.\neg X)$ from review3. The honest model has u with a PO-witness $w \in \neg X$ and an ascending tower $a_1 < a_2 < \dots$, each $a_m \in \forall \text{PO}.X$, with $\rho(a_m, w) = \text{PPI}$ for all m (machine-checked: $\text{Comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}$, $\forall \text{PO}.X$ kills PO, $\text{Comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ propagates). Here $k_2 = 1$, so the splice is $w E_{\uparrow} a_1$. The accompanying `repaired_certificate.py` builds this presentation explicitly on a depth-8 prefix and verifies:

- zero RCC5 frame-axiom violations on every triple;
- $\text{Reach}_{\text{PP}}(w, a_m)$ holds for every tower node, so the **down** shape with label PPI is available for every (a_m, w) ;
- (u, w) is residual with the inherited label PO;
- every demand of C'_0 is met, the single top existential being discharged by the blocking cycle.

Thus the repaired construction yields a valid certificate, exactly the object the round-8 construction failed to produce.

9.2 The UNSAT sibling: still correctly rejected

Let $G = (\forall \text{PO}.X) \sqcap (\forall \text{PPI}.Y) \sqcap (\exists \text{PP}.G)$ and $C_0 = (\exists \text{PP}.G) \sqcap (\exists \text{PO}.\neg X \sqcap \neg Y)$. Now $w \in \neg X \sqcap \neg Y$. The splice makes w a PPI-successor (proper part) of a_1 , so (V10) fires a_1 's $\forall \text{PPI}.Y$ on w , requiring $Y \in \text{tp}(w)$ — contradicting $w \in \neg Y$. The alternative residual label PO fires $\forall \text{PO}.X$ via (V12), requiring $X \in \text{tp}(w)$, contradicting $w \in \neg X$; and DR is composition-illegal ($\text{DR} \notin \text{Comp}(\text{PPI}, \text{PO})$). `repaired_certificate.py` enumerates the two composition-legal labels for (a_1, w) and confirms both clash, so no valid certificate exists and the procedure correctly returns UNSAT. The repair therefore does not over-accept: it adds the missing PP/PPI pair shapes *and* the universal-safety obligations that come with them.

10 Scope and remaining verification obligations

The repair is local to the witness-placement step; soundness, the occurrence-sensitive blocking machinery, the typed-equality quotient, and the request-closed cycle argument are untouched. We have proved monotonicity (Lemma 4.1), boundedness (Lemma 4.3), acyclicity (Lemma 5.1), inherited consistency (Lemma 5.2), and finiteness (Lemma 5.3); the algebra and two worked certificates are machine-checked. A careful referee should still verify:

- O1. **Excision coherence under replacement.** Lemma 4.3 excises a same-descriptor segment of the chain while a fixed off-chain witness w hangs below it. We argued the inherited label r to w is constant on the stretch, hence preserved; one should confirm the replacement lemma’s preconditions (boundary descriptor identity) genuinely subsume the labels to spliced-in witnesses crossing the boundary, i.e. that the descriptor as defined records enough to license the excision in the presence of splices. This is the single most load-bearing point.
- O2. **Simultaneous splices.** Lemmas 5.2 and 5.3 treat the splices collectively via the embedding into $\mathcal{J}$, but the interaction of many splices with the blocking decomposition (which splice lands in the prefix vs. in a recurrent profile, per Remark 4.4) deserves a fully spelled-out normal form.
- O3. **Overlap amalgam closure.** Lemma 6.1 reduces the shared-part case to existing split copies; one should check that the support-closed mosaic family (companion Definition “support-closed”) remains closed when a side witness is simultaneously a backbone node and a PO-relative of its former source.
- O4. **No interaction with the round-8 finiteness constants.** The descriptor blow-up factor in Lemma 5.3 should be carried through the explicit bounds of the round-8 enumeration appendix to confirm the universe is still effectively enumerable (it is elementary, but the constant changes).

None of these is an obstruction in the sense of D-1: they are bookkeeping obligations on a construction that, by Lemma 5.2, presents a fixed consistent model and therefore cannot violate composition or universal safety. We accordingly assess the decidability theorem as *true and provable along this route*, with the present note supplying the missing completeness argument modulo the bookkeeping items O1–O4.

Reproducibility

- `rcc5_compose.py` — round-8 composition table; D-1 forcing.
- `fix_feasibility.py` — Lemma 4.1 (monotone/absorbing transitions) and the bounded-threshold rationale.
- `repaired_certificate.py` — explicit valid repaired certificate for C'_0 (frame axioms, `Reachpp` tags, concept truth) and the correct rejection of the UNSAT sibling.

Disclaimer

Produced by Claude (Anthropic, Opus 4.8), prompted by Michael Wessel. Not independently verified by human experts. The composition arithmetic and the two worked certificates are machine-checked; the structural lemmas (4.1–6.1) are proved here but not formally verified, and obligations O1–O4 are stated as open bookkeeping.